Question 1 2 marks 101

A wheel has a diameter of 20 cm and rotates through a central angle of 252°.

Determine how far the wheel rolled.

Question 2 3 marks 102

Solve the following equation over the interval $[0,\ 2\pi]$:

$$3\sin^2\theta - 10\sin\theta - 8 = 0$$

Question 3 — 3 marks 103

Determine and simplify the fourth term in the expansion of $\left(2x^4 - 3y\right)^8$.

Question 4 — 3 marks 104

Sheeva's bank is lending her \$50 000 at an annual interest rate of 6%, compounded monthly, to purchase a car.

Given that the last payment will be a partial payment, determine how many full monthly payments of \$800 Sheeva will have to make.

The formula below may be used.

$$PV = \frac{R\left[1-(1+i)^{-n}\right]}{i}$$

where $PV =$ the present value of the amount borrowed

$R =$ the amount of each periodic payment

$$i = \frac{\text{annual interest rate (as a decimal)}}{\text{the number of compounding periods per year}}$$

$n =$ the number of equal periodic payments

Express your answer as a whole number.

Question 5 2 marks 105

An employee asked 10 people in an ice cream shop to wait in line.

Determine the number of different arrangements possible if two of the people, Jamie and John, refused to stand next to each other in the line.

Note: A calculator is not required for the remaining test questions.

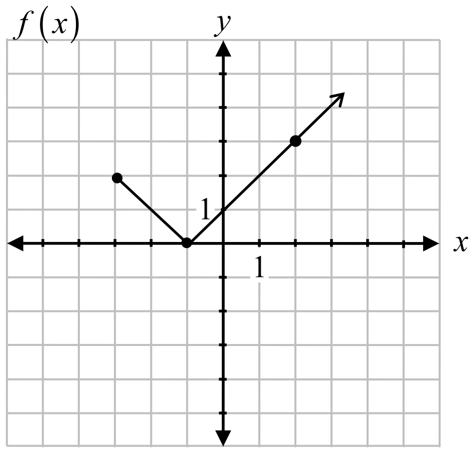
Question 6 1 mark 106

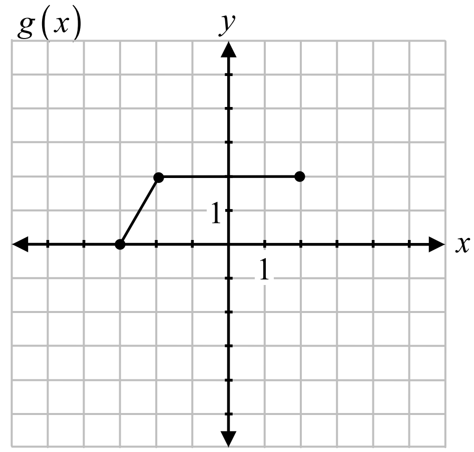
The point $(-2, 4)$ is on the graph of $f(x)$.

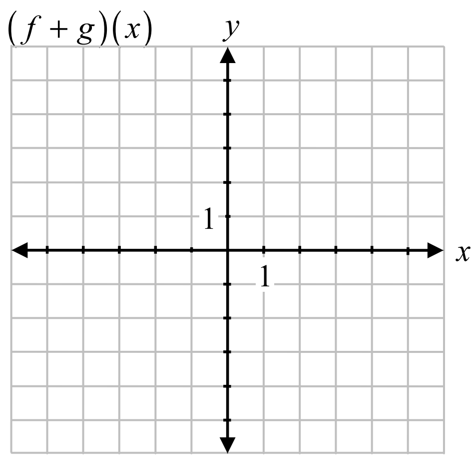
State the coordinates of the corresponding point when $f(x)$ is reflected over the y-axis.

Question 7 — 2 marks

Given the graphs of $f(x)$ and $g(x)$, sketch the graph of $(f+g)(x)$.







Question 8 3 marks

108

Using the laws of logarithms, fully expand the expression:

$$\log_2\left(\frac{w^3x}{y-1}\right)$$

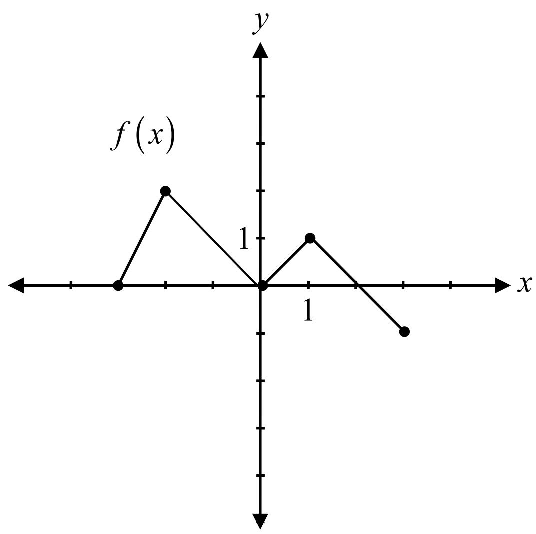
Question 9 4 marks

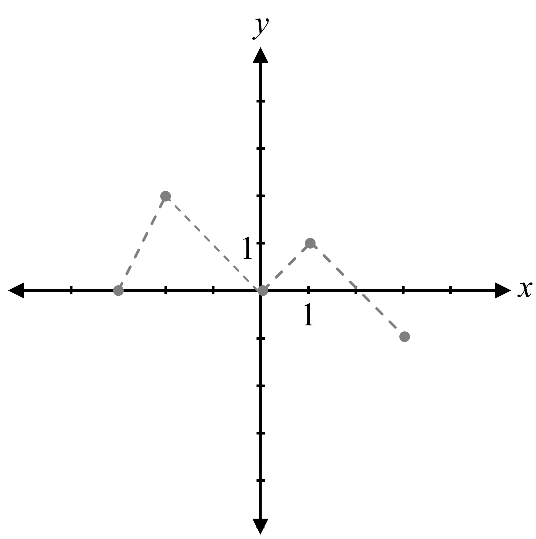
Solve the following equation algebraically for θ, where $0 \leq \theta \leq 2\pi$:

$$2\cos 2\theta = 1$$

Question 10 3 marks

Given the graph of $y = f(x)$, sketch the graph of $y = 2\left|f(x-1)\right|$.





The graph of $f(x)$ has already been drawn for your reference.

No marks will be awarded for the graph of $f(x)$.

Question 11 — 3 marks 111

Prove the identity for all permissible values of θ:

$$\cos\theta + \tan\theta\sin\theta = \frac{\tan\theta\sin\theta}{1-\cos^2\theta}$$

Left-Hand Side	Right-Hand Side

Question 12 1 mark 112

Raoul has 8 shirts, 5 pairs of pants, and 3 hats. He adds the options together and determines that he has 16 different outfits to wear.

Raoul made an error in calculating the number of different outfits. Describe how to determine the correct number of outfits.

Question 13 a) 1 mark b) 1 mark 113 114

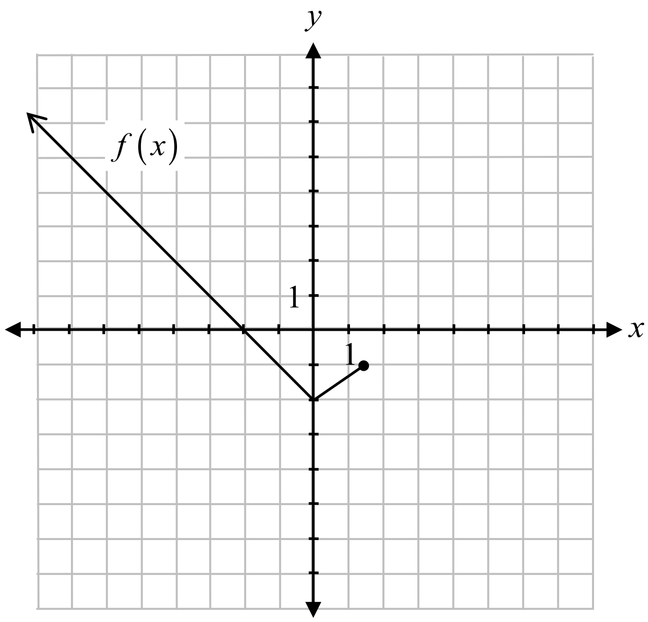
Given $f(x) = 2x - 1$ and $g(x) = x^2 + 1$:

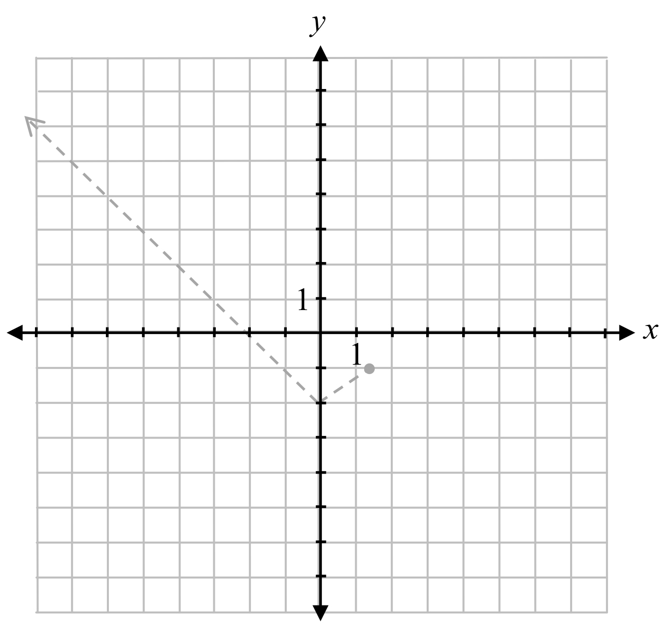
a) Determine $f(x) \cdot g(x)$.

b) Determine $g(g(x))$.

Question 14 — 2 marks

Given the graph of $y = f(x)$, sketch the graph of $y = \sqrt{f(x)}$.





The graph of $f(x)$ has already been drawn for your reference.

No marks will be awarded for the graph of $f(x)$.